General equation of two straight lines

Q.1

Choose the correct answer

The equation $ax^2 + b\,y^2 + cx + cy = 0,\ \ c \neq 0$ represents two straight lines if

a) $a + b = 0$ b) $b + c = 0$ c) $a + c = 0$ d) none of these

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If the sum of the slopes of the straight lines given $4x^2 + 2kxy - 7y^2 = 0$ is equal to the product of the slopes the k=

a) -4 b) 4 c) - 2 d) 2

If the equation $(h - c)x^2 + h\,xy + (h + c)y^2 = 0$ represents pair of straight lines which are parallel to each other, then h^2 is equal to

a) $\frac{4}{3}\,c^2$ b) $\frac{3}{4}\,c^2$ c) 0 d) None of these

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If the equation $x^2 + 4\,xy + (1 + \lambda)\,y^2 = 0$ represents two perpendicular lines, then the value of λ is

a) -1 b) $-\ 2$ c) 0 d) None of these

Q.2

Prove that each of the following equations represents two straight lines and find them

$9\,x^2 - y^2 = 0$

$4\,x^2 - 5x + y^2 = 0$

$3\,x^2 - 7xy + 2y^2 + 2y = 0$

Q.3

Find the value of k that makes each of the following equations represent two straight lines

$x^2 - xy + ky^2 - 3x - 3y = 0$

$2x^2 + k\,xy - 2y^2 - 10x + 5y = 0$